NAME – SUBASH G **COURSE NAME – OBJECT ORIENTED PRGRAMMING**
REG NO – 192572150 **COURSE CODE – DSA0108**

CO2-AT2 -SCENARIO BASED

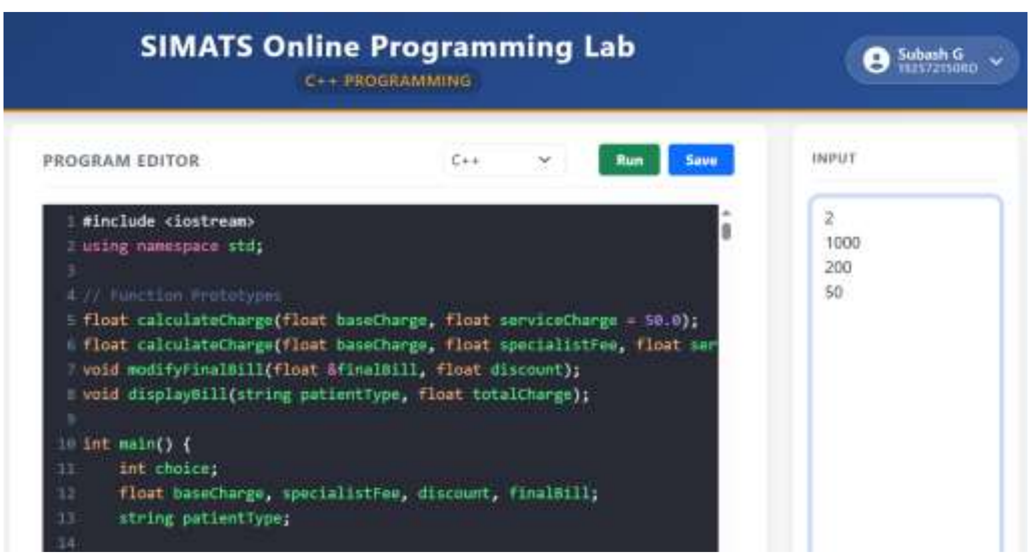
QUESTION 1 :

PROGRAM:

```
#include <iostream>
using namespace std;

int main()
{
   int choice;
   float baseCharge, specialistFee, serviceCharge, discount, total;

   cout << "===== Hospital Consultation Billing System =====" << endl;
   cout << "1. General Consultation" << endl;
   cout << "2. Specialist Consultation" << endl;

   cout << "\nEnter your choice: ";
   cin >> choice;

   cout << "Enter base consultation charge: ";
   cin >> baseCharge;

   if(choice == 2)
   {
      cout << "Enter specialist fee: ";
      cin >> specialistFee;
   }
   else
   {
      specialistFee = 0;
```

```
    }

    cout << "Enter service charge: ";
    cin >> serviceCharge;

    cout << "Enter discount amount (if any): ";
    cin >> discount;

    total = baseCharge + specialistFee + serviceCharge - discount;

    cout << "\n----- Consultation Bill -----" << endl;
    cout << "Patient Type: ";

    if(choice == 1)
        cout << "General Patient" << endl;
    else if(choice == 2)
        cout << "Specialist Patient" << endl;
    else
        cout << "Invalid Choice" << endl;

    cout << "Total Consultation Charge: Rs. " << total << endl;

    return 0;
}
```

SAMPLE INPUT:



OUTPUT:

OUTPUT

```
===== Hospital Consultation Charge System =====
1. General Consultation
2. Specialist Consultation
Enter your choice:
Enter base consultation charge:
Enter specialist fee:
Enter service charge:
Enter discount amount (if any):
```

```
----- Consultation Bill -----
Patient Type:
Specialist Patient
Total Consultation Charge: Rs. 1250
```

QUESTION 2 :

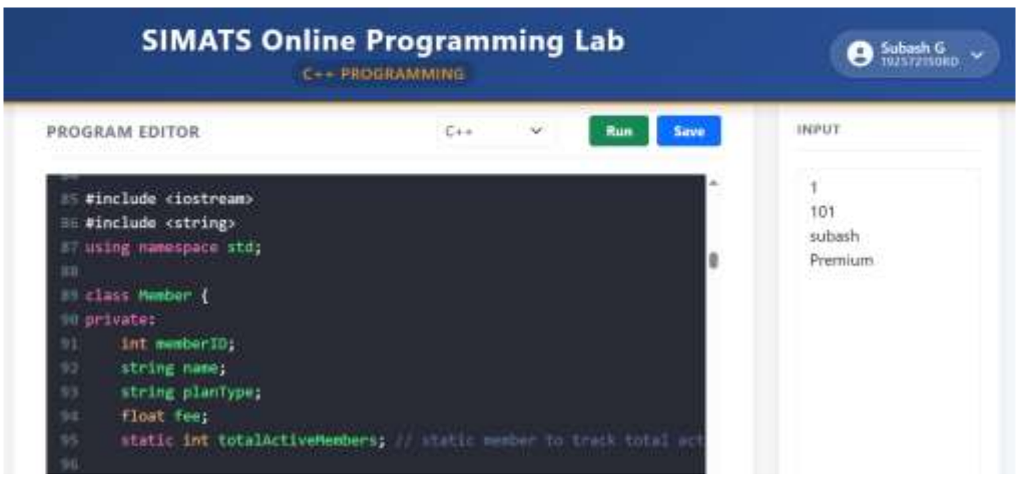
PROGRAM:

```
#include <iostream>
#include <string>
using namespace std;

struct Member
{
    int memberID;
    string name;
    string planType;
    int fee;
};

int main()
{
```

```cpp
int n;

cout << "===== Fitness Centre Membership System =====" << endl;

cout << "Enter number of members to register: ";
cin >> n;

Member m[n];

for(int i = 0; i < n; i++)
{
    cout << "\nEnter details for Member " << i + 1 << ":" << endl;

    cout << "Member ID: ";
    cin >> m[i].memberID;

    cout << "Name: ";
    cin >> m[i].name;

    cout << "Plan Type (Basic/Standard/Premium): ";
    cin >> m[i].planType;

    if(m[i].planType == "Basic" || m[i].planType == "basic")
        m[i].fee = 500;
    else if(m[i].planType == "Standard" || m[i].planType == "standard")
        m[i].fee = 1000;
    else if(m[i].planType == "Premium" || m[i].planType == "premium")
        m[i].fee = 1500;
    else
        m[i].fee = 0;
}

cout << "\n===== All Membership Details =====" << endl;

for(int i = 0; i < n; i++)
{
    cout << "\n----- Membership Details -----" << endl;
```

```
        cout << "Member ID   : " << m[i].memberID << endl;
        cout << "Name        : " << m[i].name << endl;
        cout << "Plan Type   : " << m[i].planType << endl;
        cout << "Fee         : Rs. " << m[i].fee << endl;
    }

    return 0;
}
```

SAMPLE INPUT:



OUTPUT:

OUTPUT

```
===== Fitness Centre Membership System =====
Enter number of members to register:
Enter details for Member 1:
Member ID: Name: Plan Type (Basic/Standard/Premium):
===== All Membership Details =====
----- Membership Details -----
Member ID   : 101
Name        : subash
Plan Type   : Premium
Fee         : Rs. 1500
```

QUESTION 3:

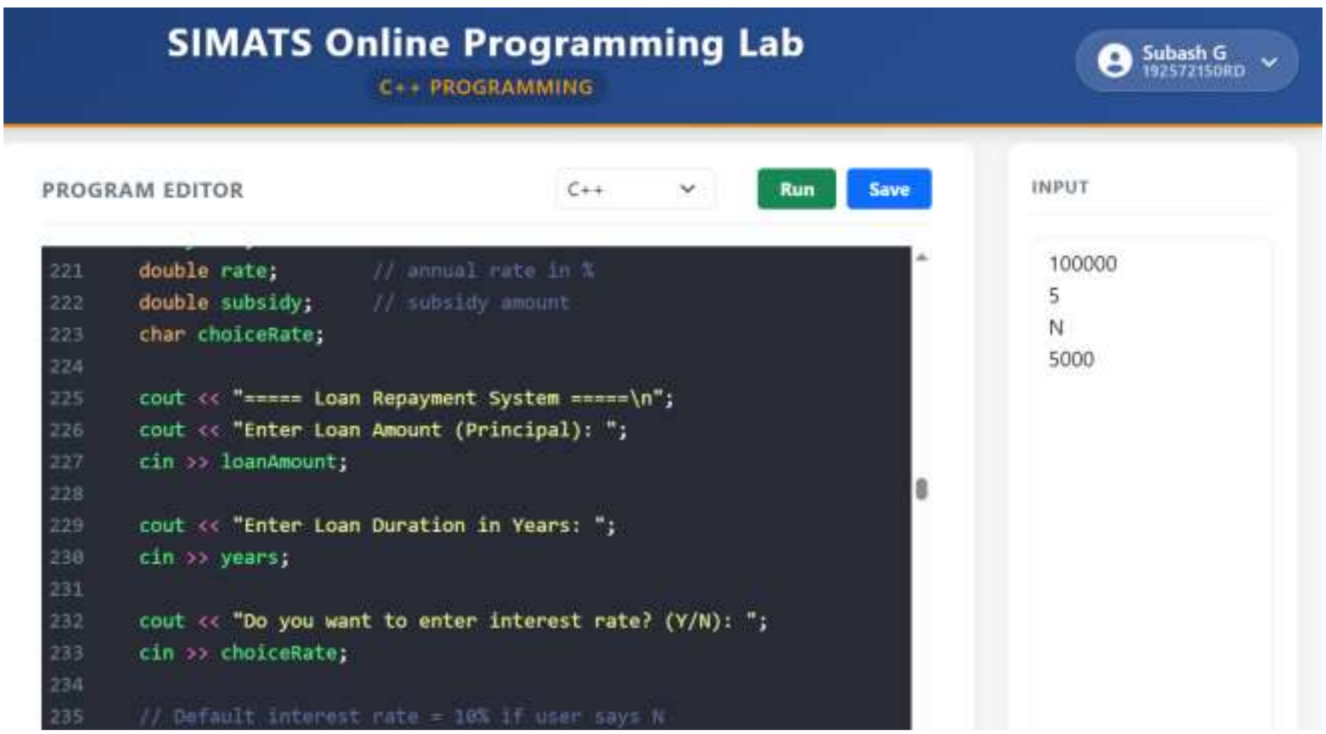
PROGRAM:

```
#include <iostream>

#include <iomanip>

#include <cmath>

using namespace std;


// Function to calculate EMI

double calculateEMI(double principal, double annualRate, int years)

{

    double monthlyRate = annualRate / (12 * 100);

    int months = years * 12;


    double emi = (principal * monthlyRate * pow(1 + monthlyRate, months)) /

                 (pow(1 + monthlyRate, months) - 1);


    return emi;

}

int main()

{

    double principal, subsidy = 0, payableAmount;

    double interestRate = 10.0;

    char choice;

    int years;

    cout << "===== Loan Repayment System =====" << endl;
```

```
cout << "Enter Loan Amount (Principal): ";
cin >> principal;
cout << "Enter Loan Duration in Years: ";
cin >> years;

cout << "Do you want to enter interest rate? (Y/N): ";
cin >> choice;

if(choice == 'Y' || choice == 'y')
{
   cout << "Enter Annual Interest Rate (%): ";
   cin >> interestRate;
}
cout << "Enter Subsidy Amount (if any): ";
cin >> subsidy;
payableAmount = principal - subsidy;
double emi = calculateEMI(payableAmount, interestRate, years);
int months = years * 12;
double totalRepayment = emi * months;
double totalInterest = totalRepayment - payableAmount;
// Simple Interest (Reference)
double simpleInterest = (payableAmount * interestRate * years) / 100;
cout << fixed << setprecision(2);
cout << "\n===== Repayment Details =====" << endl;
cout << "Original Loan Amount: Rs. " << principal << endl;
```

```
    cout << "Subsidy Provided:     Rs. " << subsidy << endl;
    cout << "Payable Loan Amount:  Rs. " << payableAmount << endl;
    cout << "Annual Interest Rate: " << interestRate << " %" << endl;
    cout << "Loan Duration:        " << years << " years (" << months << " months)" << endl;
    cout << "\nMonthly EMI:         Rs. " << emi << endl;
    cout << "Total Interest:       Rs. " << totalInterest << endl;
    cout << "Total Repayment:      Rs. " << totalRepayment << endl;
    cout << "\nSimple Interest (for reference): Rs. " << simpleInterest << endl;
    return 0;
}
```

SAMPLE INPUT:



OUTPUT:

OUTPUT

```
===== Loan Repayment System =====
Enter Loan Amount (Principal): Enter Loan Duration in Years: Do
you want to enter interest rate? (Y/N): Enter Subsidy Amount (if
any):
===== Repayment Details =====
Original Loan Amount: Rs. 100000.00
Subsidy Provided:      Rs. 5000.00
Payable Loan Amount:  Rs. 95000.00
Annual Interest Rate: 10.00 %
Loan Duration:        5 years (60 months)
```

OUTPUT

```
Payable Loan Amount:  Rs. 95000.00
Annual Interest Rate: 10.00 %
Loan Duration:        5 years (60 months)

Monthly EMI:          Rs. 2018.47
Total Interest:      Rs. 26108.15
Total Repayment:     Rs. 121108.15

Simple Interest (for reference): Rs. 47500.00
```

QUESTION 4:

PROGRAM:

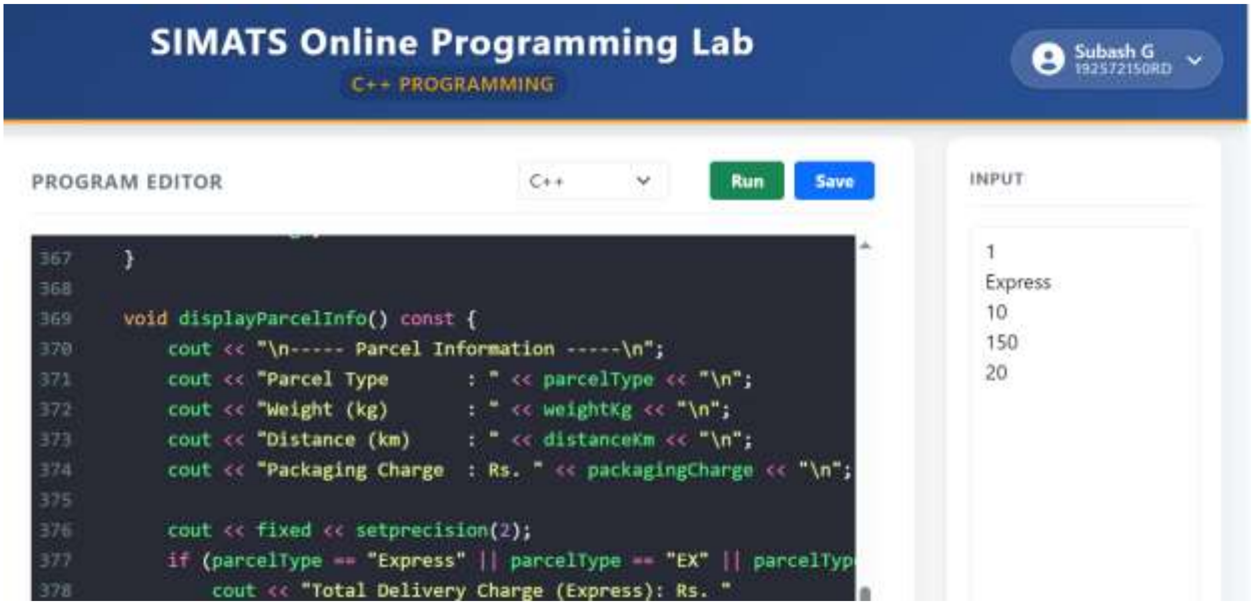
```
#include <iostream>
#include <iomanip>
#include <string>
using namespace std;

class Parcel
{
```

```
public:
    string type;
    double weight;
    double distance;
    double packagingCharge;
    double totalCharge;

    void getDetails()
    {
        cout << "Enter Parcel Type (Normal/Express): ";
        cin >> type;

        cout << "Enter Weight (kg): ";
        cin >> weight;

        cout << "Enter Distance (km): ";
        cin >> distance;

        cout << "Enter Packaging Charge (or -1 to use default 50): ";
        cin >> packagingCharge;

        if(packagingCharge == -1)
            packagingCharge = 50;
    }
```

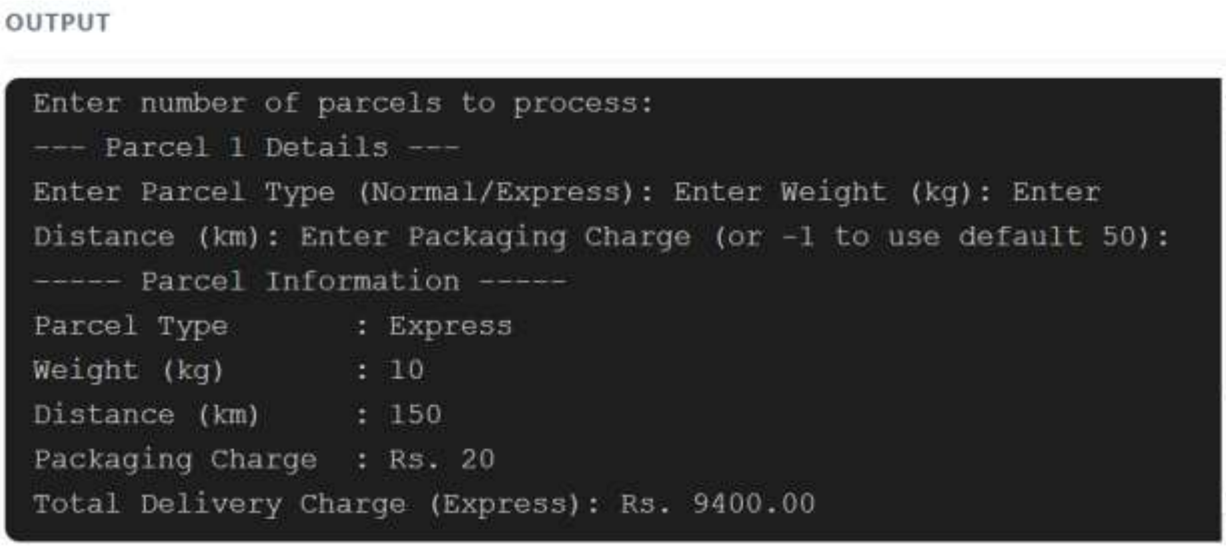
```
    void calculateCharge()
    {
        if(type == "Express" || type == "express")
        {
            totalCharge = (weight * distance * 6.25) + packagingCharge;
        }
        else
        {
            totalCharge = (weight * distance * 3.5) + packagingCharge;
        }
    }

    void display()
    {
        cout << "\n----- Parcel Information -----" << endl;
        cout << "Parcel Type       : " << type << endl;
        cout << "Weight (kg)       : " << weight << endl;
        cout << "Distance (km)     : " << distance << endl;
        cout << "Packaging Charge  : Rs. " << packagingCharge << endl;
        cout << fixed << setprecision(2);
        cout << "Total Delivery Charge (" << type << "): Rs. "
             << totalCharge << endl;
    }
};
```

```
int main()
{
    int n;

    cout << "Enter number of parcels to process: ";
    cin >> n;

    Parcel p[n];

    for(int i = 0; i < n; i++)
    {
        cout << "\n--- Parcel " << i + 1 << " Details ---" << endl;

        p[i].getDetails();
        p[i].calculateCharge();
    }

    for(int i = 0; i < n; i++)
    {
        p[i].display();
    }

    return 0;
}
```

SAMPLE INPUT:



OUTPUT:



QUESTION 5:

PROGRAM:

#include <iostream>

#include <iomanip>

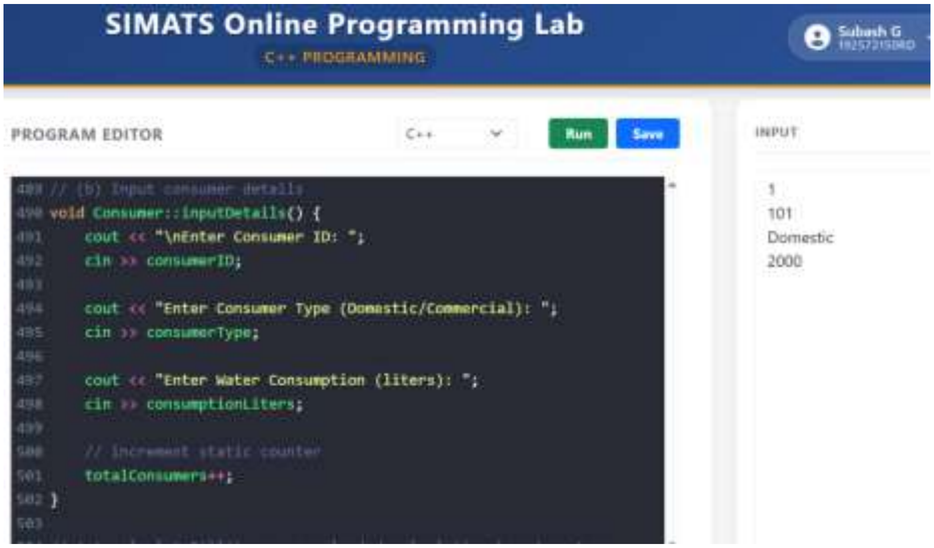
```
#include <string>
using namespace std;

class WaterBill
{
private:
    int consumerID;
    string consumerType;
    double consumption;
    double baseBill;
    double finalBill;
    double serviceTax;
    double totalPayable;
public:
    void getDetails()
    {
        cout << "Enter Consumer ID: ";
        cin >> consumerID;
        cout << "Enter Consumer Type (Domestic/Commercial): ";
        cin >> consumerType;
        cout << "Enter Water Consumption (liters): ";
        cin >> consumption;
    }
    void calculateBill()
    {
```

```
{
    // Rs.0.02 per liter
    baseBill = consumption * 0.02;
    // Subsidy
    if(consumerType == "Domestic" || consumerType == "domestic")
        finalBill = baseBill * 0.80;     //20% subsidy
    else
        finalBill = baseBill;

    //10% Service Tax
    serviceTax = finalBill * 0.10;

    totalPayable = finalBill + serviceTax;
}
void displaySummary()
{
    cout << left
         << setw(12) << consumerType
         << setw(15) << consumption
         << "Subsidy-Adjusted Bill: Rs. "
         << fixed << setprecision(2)
         << finalBill << endl;
}

void displayDetails()
{
```

```
        cout << fixed << setprecision(2);

        cout << "\n===== Detailed Billing (Default Service Tax Applied) =====\n";

        cout << "\n===== Billing Information =====\n";

        cout << "Consumer ID            : " << consumerID << endl;
        cout << "Consumer Type          : " << consumerType << endl;
        cout << "Water Consumption      : " << consumption << " liters" << endl;
        cout << "Base Bill Amount       : Rs. " << baseBill << endl;
        cout << "Final Bill(after subsidy): Rs. " << finalBill << endl;
        cout << "Service Tax (10.00%)   : Rs. " << serviceTax << endl;
        cout << "Total Payable          : Rs. " << totalPayable << endl;
    }
};

int main()
{
    int n;

    cout << "Water Billing System\n\n";

    cout << "Enter number of consumers to process: ";
    cin >> n;
```

```
    WaterBill consumer[n];


    for(int i = 0; i < n; i++)
    {
        cout << "\n--- Consumer " << i + 1 << " ---\n\n";


        consumer[i].getDetails();
        consumer[i].calculateBill();
    }
    // Summary
    cout << "\n===== Consumer Summaries (After Subsidy) =====\n";


    for(int i = 0; i < n; i++)
    {
        consumer[i].displaySummary();
    }
    // Detailed Bills
    for(int i = 0; i < n; i++)
    {
        consumer[i].displayDetails();
    }
    return 0;
}
```

SAMPLE INPUT:



OUTPUT: